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Introduction

This report focuses on advanced Security Operations Center (SOC) concepts including advanced log analysis, threat intelligence integration, incident escalation workflows, alert triage, evidence preservation, and SOC incident handling. The purpose of this task was to understand how cybersecurity analysts investigate threats, correlate logs, validate indicators of compromise, escalate incidents, and preserve evidence in a structured SOC environment.

Advanced Log Analysis

Advanced log analysis is an important SOC activity used to identify suspicious activities and correlate security events from multiple sources.

Core Concepts

Log correlation helps analysts connect events from multiple systems such as firewall logs, endpoint logs, and authentication logs to identify attack patterns.

Anomaly detection helps identify unusual behaviors such as suspicious login attempts, abnormal network activity, or unexpected outbound communication.

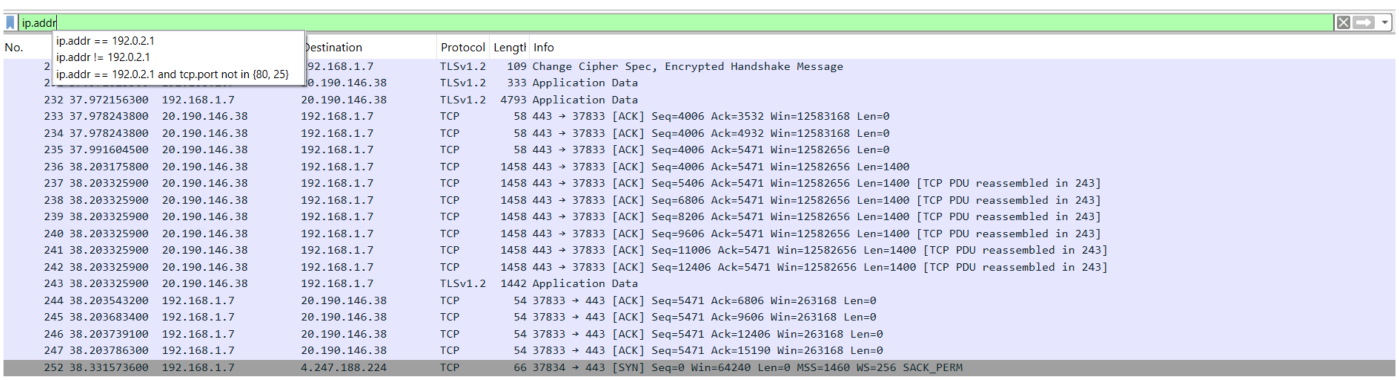
Log enrichment improves visibility by adding contextual information such as IP geolocation, user details, and system metadata.

Log Correlation Example

Timestamp	Event ID	Source IP	Destination IP	Notes
Example	4625	192.168.1.100	8.8.8.8	Suspicious DNS request

Practical Activity

Network traffic was analyzed using Wireshark to understand communication behavior and identify suspicious activity patterns.



Learning from this Task

Learned how SOC analysts correlate logs from different sources to detect suspicious behavior and reduce false positives during investigations.

Threat Intelligence Integration

Threat intelligence helps organizations identify malicious indicators and improve threat detection.

Core Concepts

Indicators of Compromise (IOCs) such as malicious IP addresses, suspicious hashes, and abnormal domains are used to detect threats.

MITRE ATT&CK techniques help analysts map attacker behavior to known tactics and procedures.

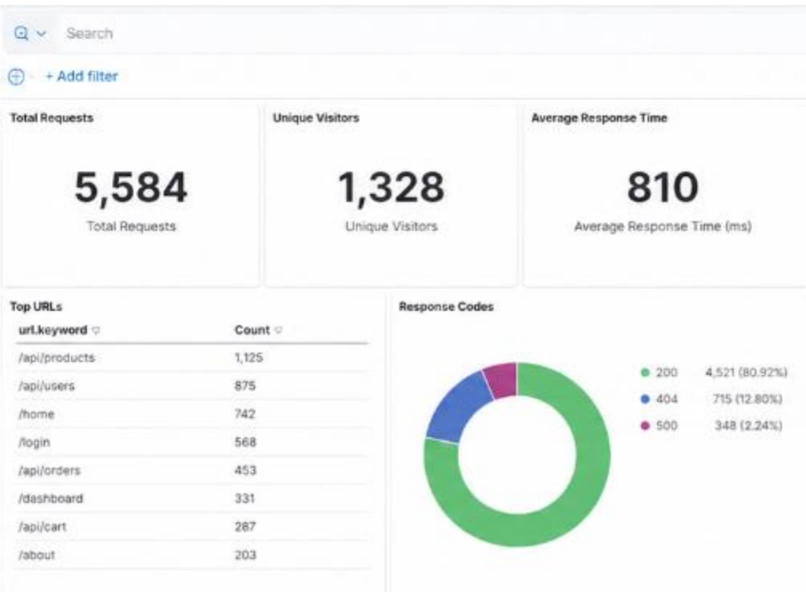
Threat feeds such as AlienVault OTX provide intelligence regarding malicious infrastructure and known attack indicators.

Threat Intelligence Example

Alert ID	IP Address	Reputation	Notes
003	192.168.1.100	Suspicious	Linked to abnormal activity

Practical Activity

Threat intelligence concepts were studied to understand how suspicious IPs and attack indicators are validated using external intelligence feeds.



Learning from this Task

Learned how threat intelligence helps enrich alerts and improves the identification of malicious activities in SOC environments.

Incident Escalation Workflows

Incident escalation workflows are used to ensure security incidents are investigated by the appropriate SOC team based on severity and complexity.

SOC Escalation Structure

Tier 1 Analysts are responsible for alert monitoring and initial triage.

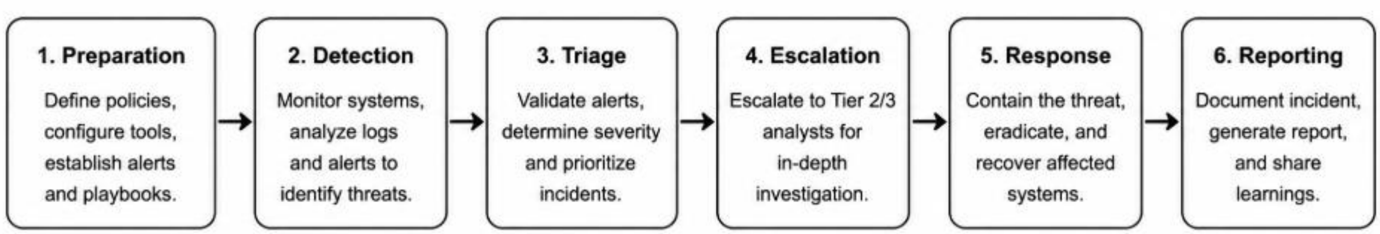
Tier 2 Analysts perform deeper investigations and incident analysis.

Tier 3 Analysts handle advanced threats and malware investigations.

SOC Managers coordinate incident handling and communication.

Escalation Workflow

Preparation → Detection → Triage → Escalation → Investigation → Containment → Recovery → Reporting



Learning from this Task

Learned how security incidents are escalated across SOC teams to improve incident handling and investigation efficiency.

Alert Triage with Threat Intelligence

Alert triage is used to determine whether suspicious activities represent real threats or false positives.

Alert Analysis Example

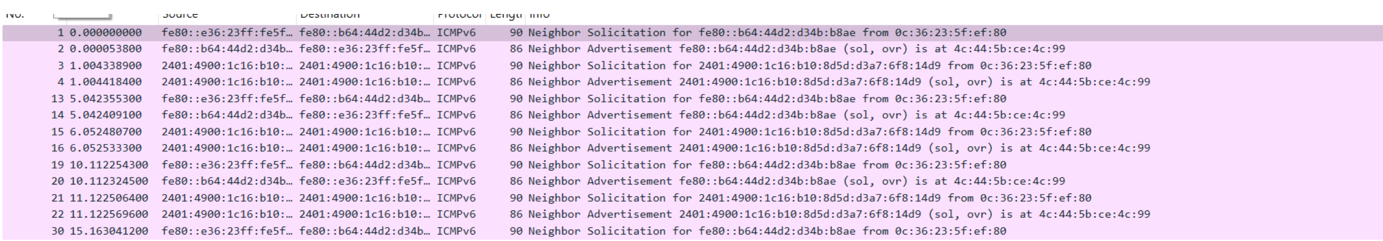
Alert ID	Description	Source IP	Priority	Status
004	Suspicious PowerShell Activity	192.168.1.101	High	Open

Threat Validation

Threat intelligence sources help validate suspicious activities and identify malicious indicators.

Practical Activity

ICMP traffic was analyzed using Wireshark to observe repeated communication patterns and identify unusual activity.



1	0.000000000	fe80::e36:23ff:fe5f…	fe80::b64:44d2:d34b…	ICMPv6	90	Neighbor Solicitation for fe80::b64:44d2:d34b:b8ae from 0c:36:23:5f:ef:80
2	0.000053800	fe80::b64:44d2:d34b…	fe80::e36:23ff:fe5f…	ICMPv6	86	Neighbor Advertisement fe80::b64:44d2:d34b:b8ae (sol, ovr) is at 4c:44:5b:ce:4c:99
3	1.004338900	2401:4900:1c16:b10:…	2401:4900:1c16:b10:…	ICMPv6	90	Neighbor Solicitation for 2401:4900:1c16:b10:8d5d:d3a7:6f8:14d9 from 0c:36:23:5f:ef:80
4	1.004418400	2401:4900:1c16:b10:…	2401:4900:1c16:b10:…	ICMPv6	86	Neighbor Advertisement 2401:4900:1c16:b10:8d5d:d3a7:6f8:14d9 (sol, ovr) is at 4c:44:5b:ce:4c:99
13	5.042355300	fe80::e36:23ff:fe5f…	fe80::b64:44d2:d34b…	ICMPv6	90	Neighbor Solicitation for fe80::b64:44d2:d34b:b8ae from 0c:36:23:5f:ef:80
14	5.042409100	fe80::b64:44d2:d34b…	fe80::e36:23ff:fe5f…	ICMPv6	86	Neighbor Advertisement fe80::b64:44d2:d34b:b8ae (sol, ovr) is at 4c:44:5b:ce:4c:99
15	6.052480700	2401:4900:1c16:b10:…	2401:4900:1c16:b10:…	ICMPv6	90	Neighbor Solicitation for 2401:4900:1c16:b10:8d5d:d3a7:6f8:14d9 from 0c:36:23:5f:ef:80
16	6.052533300	2401:4900:1c16:b10:…	2401:4900:1c16:b10:…	ICMPv6	86	Neighbor Advertisement 2401:4900:1c16:b10:8d5d:d3a7:6f8:14d9 (sol, ovr) is at 4c:44:5b:ce:4c:99
19	10.112254300	fe80::e36:23ff:fe5f…	fe80::b64:44d2:d34b…	ICMPv6	90	Neighbor Solicitation for fe80::b64:44d2:d34b:b8ae from 0c:36:23:5f:ef:80
20	10.112324500	fe80::b64:44d2:d34b…	fe80::e36:23ff:fe5f…	ICMPv6	86	Neighbor Advertisement fe80::b64:44d2:d34b:b8ae (sol, ovr) is at 4c:44:5b:ce:4c:99
21	11.122506400	2401:4900:1c16:b10:…	2401:4900:1c16:b10:…	ICMPv6	90	Neighbor Solicitation for 2401:4900:1c16:b10:8d5d:d3a7:6f8:14d9 from 0c:36:23:5f:ef:80
22	11.122569600	2401:4900:1c16:b10:…	2401:4900:1c16:b10:…	ICMPv6	86	Neighbor Advertisement 2401:4900:1c16:b10:8d5d:d3a7:6f8:14d9 (sol, ovr) is at 4c:44:5b:ce:4c:99
30	15.163041200	fe80::e36:23ff:fe5f…	fe80::b64:44d2:d34b…	ICMPv6	90	Neighbor Solicitation for fe80::b64:44d2:d34b:b8ae from 0c:36:23:5f:ef:80

Learning from this Task

Learned how SOC analysts investigate alerts, validate suspicious activities, and prioritize incidents using contextual information.

Evidence Preservation and Analysis

Evidence preservation is important for maintaining the integrity of digital investigations.

Evidence Collection Example

Item	Description	Collected By	Date	Hash Value
Memory Dump	Server-Y Analysis	SOC Analyst	Example	SHA256 Example

Importance of Evidence Preservation

Maintaining evidence ensures proper forensic investigation and prevents tampering with critical information.

Learning from this Task

Learned the importance of preserving digital evidence and maintaining proper investigation records during incident response.

Capstone Project: Full SOC Workflow Simulation

A simulated SOC workflow was studied to understand how incidents move through detection, triage, escalation, response, and reporting.

Attack Simulation

A vulnerability assessment activity was studied using penetration testing concepts and service enumeration.

Detection and Triage

Suspicious activity was identified through monitoring and analyzed to understand possible attack behavior.

Response and Containment

The affected system was isolated and suspicious activity was investigated to prevent further impact.

SOC Workflow

Detection → Alert Generation → Triage → Escalation → Investigation → Containment → Reporting

```
msf > nmap -sV 127.0.0.1
[*] exec: nmap -sV 127.0.0.1

Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-15 14:23 IST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000030s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 10.2p1 Debian 3 (protocol 2.0)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at ht
Nmap done: 1 IP address (1 host up) scanned in 0.66 seconds
```

Learning from this Task

Learned how security teams follow structured workflows to investigate incidents, contain threats, and document findings.

Overall Learnings

Learned how advanced log analysis helps identify suspicious activities through event correlation.

Understood how threat intelligence improves alert validation and threat detection.

Gained knowledge about escalation workflows, alert triage, evidence preservation, and SOC incident handling.

Challenges Faced

Some advanced SOC tools required complex configurations and setup. These challenges were overcome by studying workflow simulations, threat intelligence integration, and network traffic analysis to understand SOC processes effectively.

Conclusion

This task provided practical understanding of advanced SOC operations including log correlation, threat intelligence integration, escalation workflows, evidence preservation, and alert triage. It improved understanding of how cybersecurity teams investigate incidents, validate threats, and coordinate incident response activities in professional SOC environments.

Sources Consulted

MITRE ATT&CK Framework
AlienVault OTX Documentation
Wireshark Documentation
Cybersecurity Learning Resources
SOC Investigation References
